OPERATIONAL RESEARCH & OPTIMISATION

Dr. Fang He
Email: F.He@westminster.ac.uk
Office: N7.118
University of Westminster

Module Team

- Module leader:
  - Dr. Fang He, hef@westminster.ac.uk
- Tutors:
  - Dr. Fang He
  - Dr. Salma Chahed
  - Dr. Sheema Noorain

Overview of the Module

Aim of the Module

Achieve an understanding of operational research techniques with emphasis in computational modelling and optimisation.

Operational Research (OR) is a fascinating discipline that uses modelling techniques , analytical methods and algorithms to analyse and solve complex real-world problems. OR is the application of modelling, statistics, programming and other general problem-solving skills in improving operations and business processes.

Using OR techniques decision-makers can make more effective decisions and build more productive systems.

Weekly teaching schedule and assessments

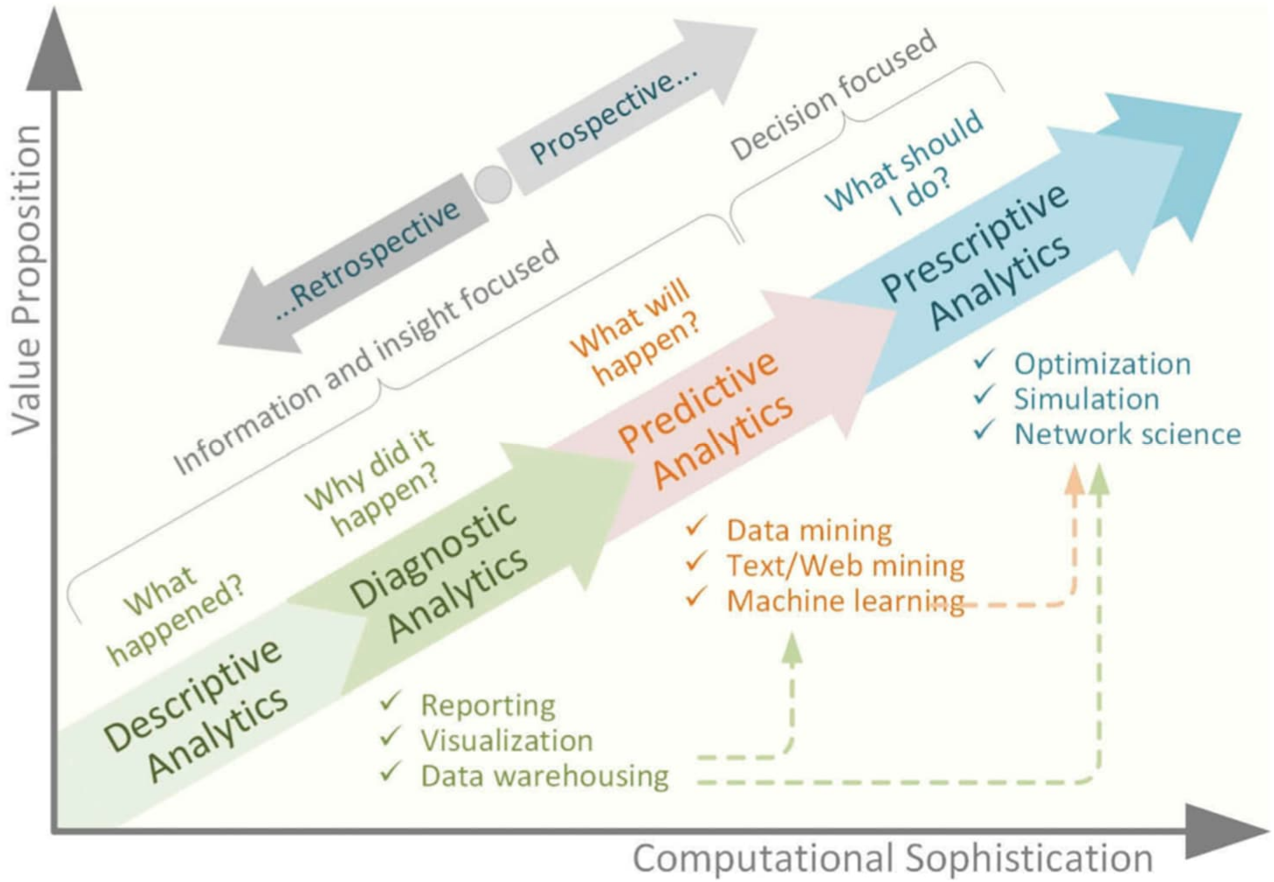
	Lecture	Tutorial	Extended reading
Week 1(19/1/2026)	Into and Linear Programming	Linear Programming	Paper: Research challenges and opportunities in business analytics
Week 2(26/1/2026)	Network model 1	Sensitivity analysis and Network model 1	Book Chapter: Mathematical Programming: An Overview
Week 3(2/2/2026)	Network model 2	Network model 2	Paper: Applications of network
Week 4(9/2/2026)	Integer Programming	Integer Programming	
Week 5(16/2/2026)	Goal programming	Goal programming	
Week 6(23/2/2026)	Reading week		
Week 7(2/3/2026)	Case study	Case study	
Week 8(9/3/2026)	Heuristics	Team project presentation	
Week 9(16/3/2026)	Markov Chain	Markov Chain	
Week 10(23/3/2026)	Markov decision process	Markov decision process	
Week 11(30/3/2026)	Game Theory	Game Theory	
Week 12(6/4/2026)	Revision lecture	ICT-MCQ (During Tutorial time, 90 minutes)	

Assessment methods and weights

Assessment name	Weighting %	Qualifying mark %	Qualifying set (where the minimum mark required applies across multiple assessments)	Assessment type (e.g. essay, presentation, open exam or closed exam)
Team project Presentation	30%	30%	Set 1	Group presentation
Team project Individual report	30%	30%	Set 1	
ICT	40%	30%	N/A	In class test

- Team project: Q&A, marking scheme etc.
- ICT: past paper.

Characterising business analytics along value proposition and computational sophistication



Dursun Delen, Research challenges and opportunities in business analytics, 2018

Introduction to Optimisation

- “Optimsation” comes from the same root as “optimal”, which means *best*. When you optimise something, you are “making it best”.

Introduction to Optimisation

- Decision variables: the elements that are under the control of the decision maker.
  - The work schedules of each employee
  - The level of investments in a portfolio
  - An important factor, reflect on the problem modelling and complexity of the model.
- A single objective function (of the decision variables)
  - minimize cost or …
  - maximize expected return or …
  - Multi-objective (later on)
- Constraints: restrictions on the decision variables (level of description of the problem you want to achieve)
  - “Business rules”
    - no worker can work more than 5 consecutive days
    - There is at most 2% investment in any stock in the portfolio
  - “Physical laws”
    - No worker can work a negative amount of time .

A brainstorm question

- Optimisation is everywhere!
- Example:

A sales rep sells 12 models of vacuum cleaners. During one day he has to visit 9 customers by car to demonstrate his products.

Given data:

distances between locations

trunk size of the car and sizes of models

expected profit / utility for demonstrating each model

What kinds of optimisation problem present here?

Introduction to Optimisation

What kinds of optimisation problem present here?

Problem 1: What is the shortest route? ("Traveling Salesman problem")

Problem 2: All the 12 models of vacuum cleaners do not fit in the car. What models to take? ("Knapsack Problem")

Approach to the optimisation problem

1. Modelling: decide on the decision variables, the objective, the constraints, and how these describe the problem;

   Algebraic model: Derive correct algebraic expressions;

2. Solve: Systematically choose the values of the decision variables that make the objective as large (for maximization) or small (for minimization) as possible and cause all constraints to be satisfied.

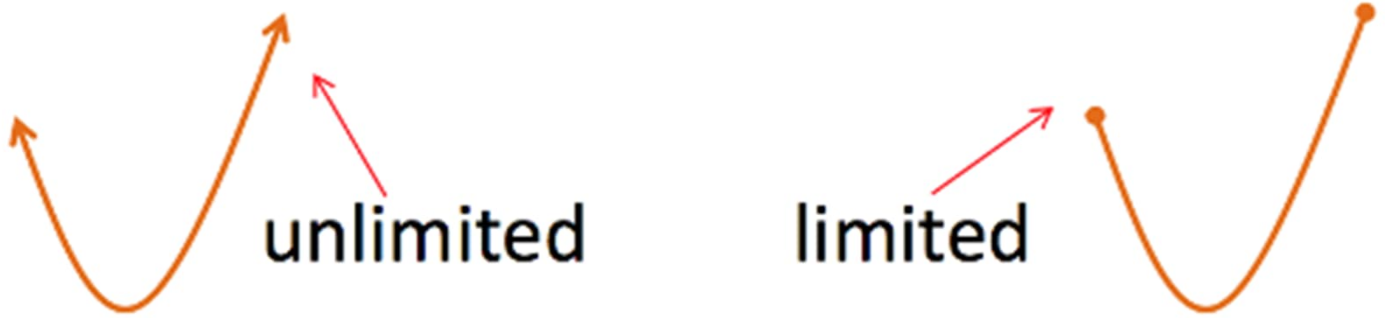
   Optimisation algorithms: Simplex, Interior-point method, Brach-and-Bound, etc. (out of scope of this module)

What we will use in the module

- **EXCEL-solver**
  - Excel—the most heavily used spreadsheet package on the market
  - Solver Add-in—uses powerful algorithms to perform spreadsheet optimisation.
- **Additional solver you may look into:**
  - **AIMMS, AMPL, OPL**: Algebraic modelling language, integrated development environment,
  - **Solvers**, such as: **CPLEX, CP optimizer**, **Gurobi**, etc.

Types of Optimisation problems

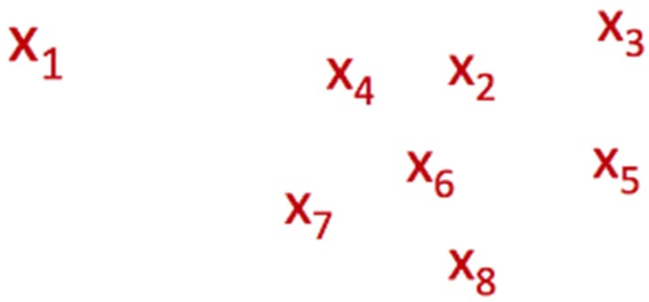
- Some problems have constraints and some do not.



- Resource scarcity: we deal with constrained optimisation in this module.

Types of Optimisation problems

- There can be one variable or many.

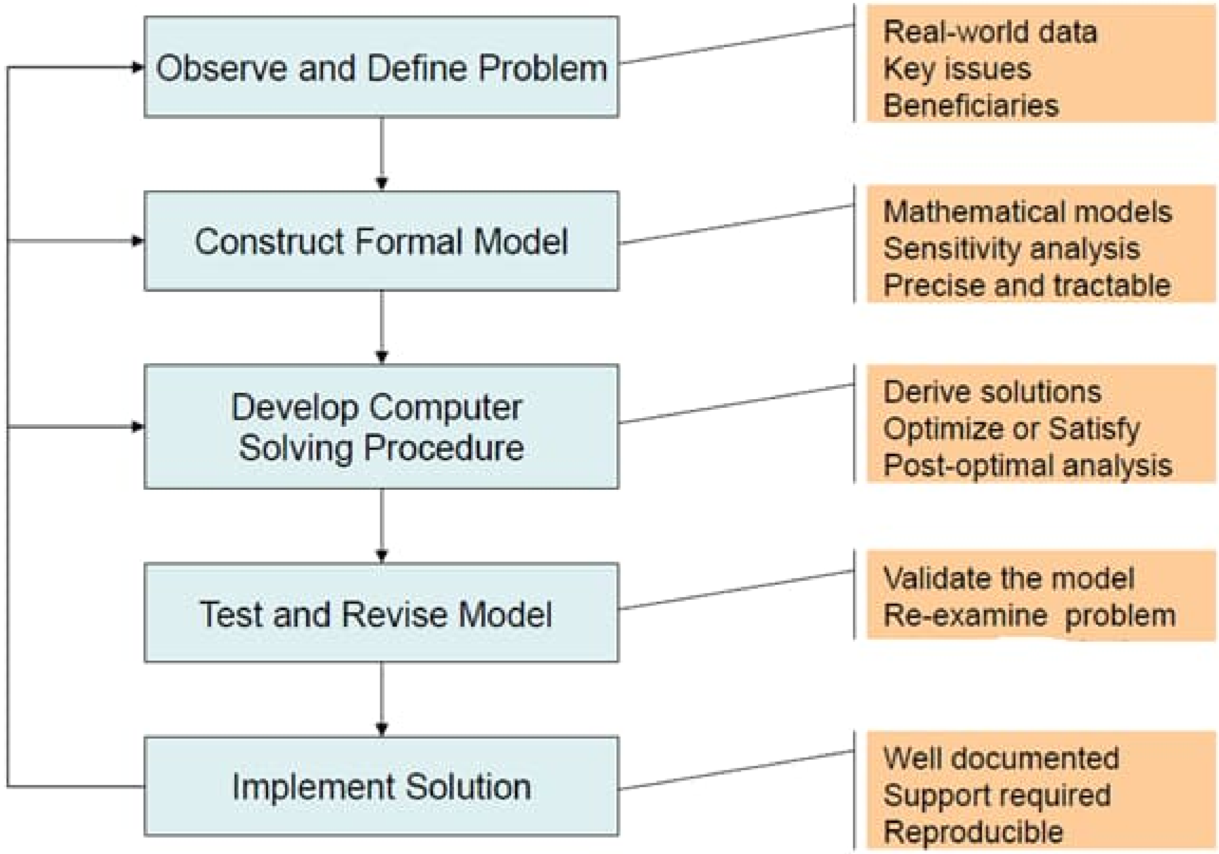


- Variables can be discrete (for example, only have integer values) or continuous.

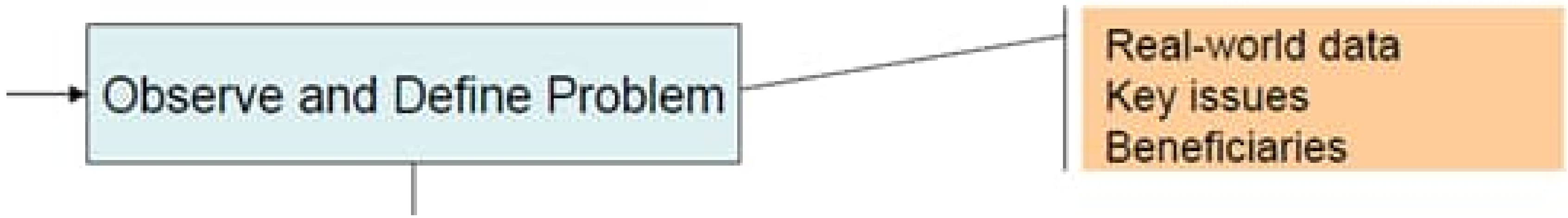
Types of Optimisation problems

- Some problems are static (do not change over time) while some are dynamic (continues adjustments must be made as changes occur)
- System can be deterministic (specific causes produce specific effects) or stochastic (involve randomness)
- We will investigate both in the module.

General framework for optimisation



1. Defining the problem and gathering data



- Advisory role;
- Vague, imprecise described practical problems➔ study relevant system and develop a well-defined statement of the problem to be considered.
- Crucial step: It is difficult to extract a “right” answer from the “wrong” problem!

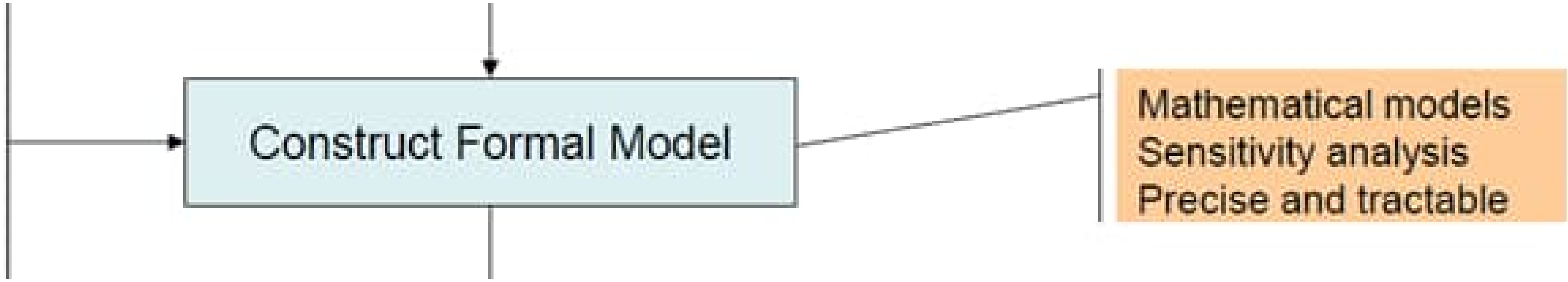
1. Defining the problem and gathering data(continue)

- Ascertaining the *appropriate objectives*
- Identify the member (or members) of management who actually will be making the decisions concerning the system
- *gathering relevant data* about the problem.
- To gain an accurate understanding of the problem
- And to provide the needed input for the mathematical model

1. Defining the problem and gathering data(continue)

- Data gathering:
  - Usually not ready to collect straightaway;
  - Information management system;
  - Estimation of the data based on historical data/forecasting/educated guess/etc.

2. Formulate a mathematical model

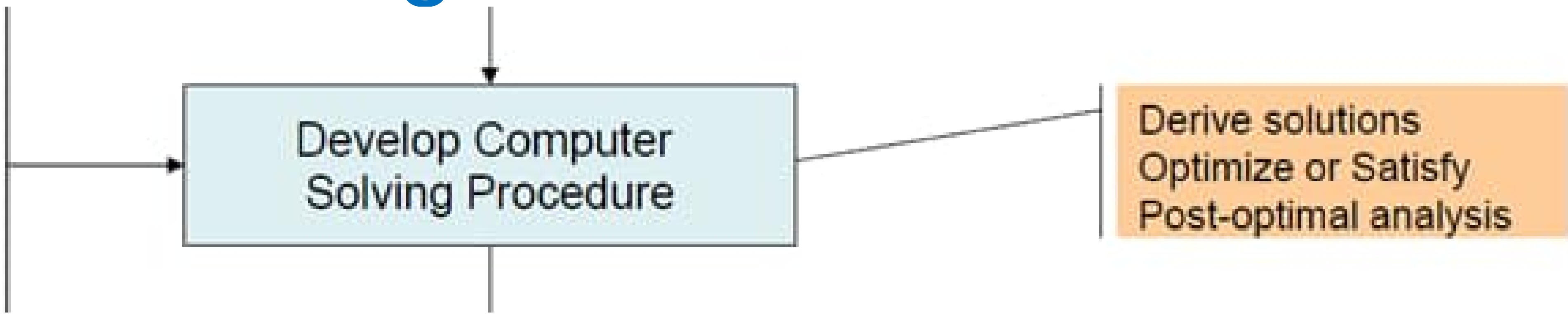


- Decision variables;
- Constraints;
- Objective function;
- Parameters: Determining the appropriate values to assign to the parameters of the model is both critical and challenging →**rough estimation**;

2. Formulate a mathematical model(continue)

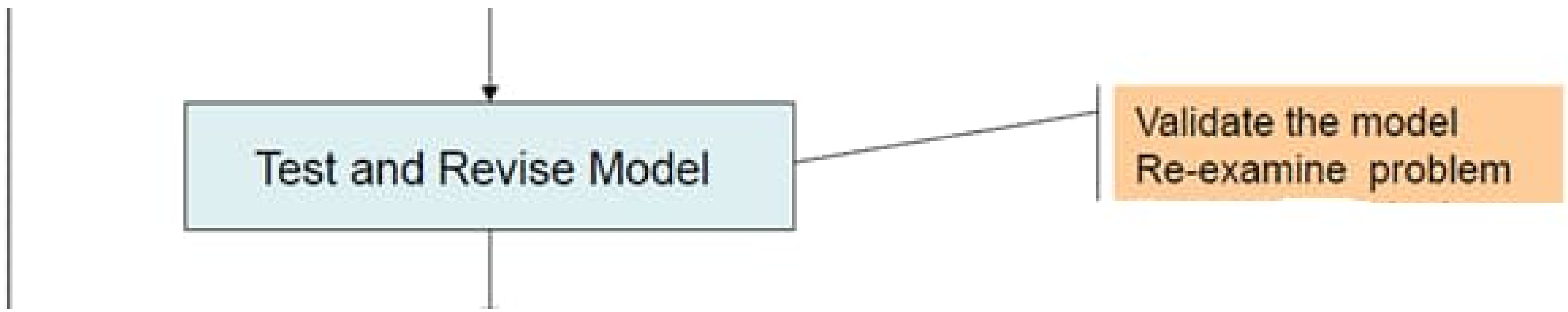
- **Precise**: Considering all its interrelationships simultaneously;
- **Tractable**: as long as a valid representation; not necessary to include unimportant details or factors;
- A good approach: Begin with a very simple version➔ evolutionarily to more elaborate models to reflect the complexity of the real problem.
- **Trade-off** between the *precision* and the *tractability* of the model.

3. Deriving solutions from model



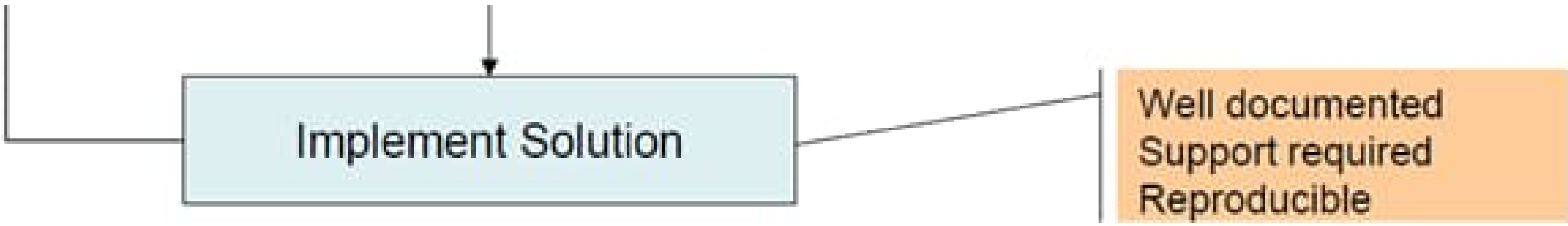
- Optimal solution;
- Satisfactory solution;
- Heuristic procedures;
- **what-if analysis:** *what* would happen to the optimal solution *if* different assumptions are made about future conditions.

4. Test and revise the model



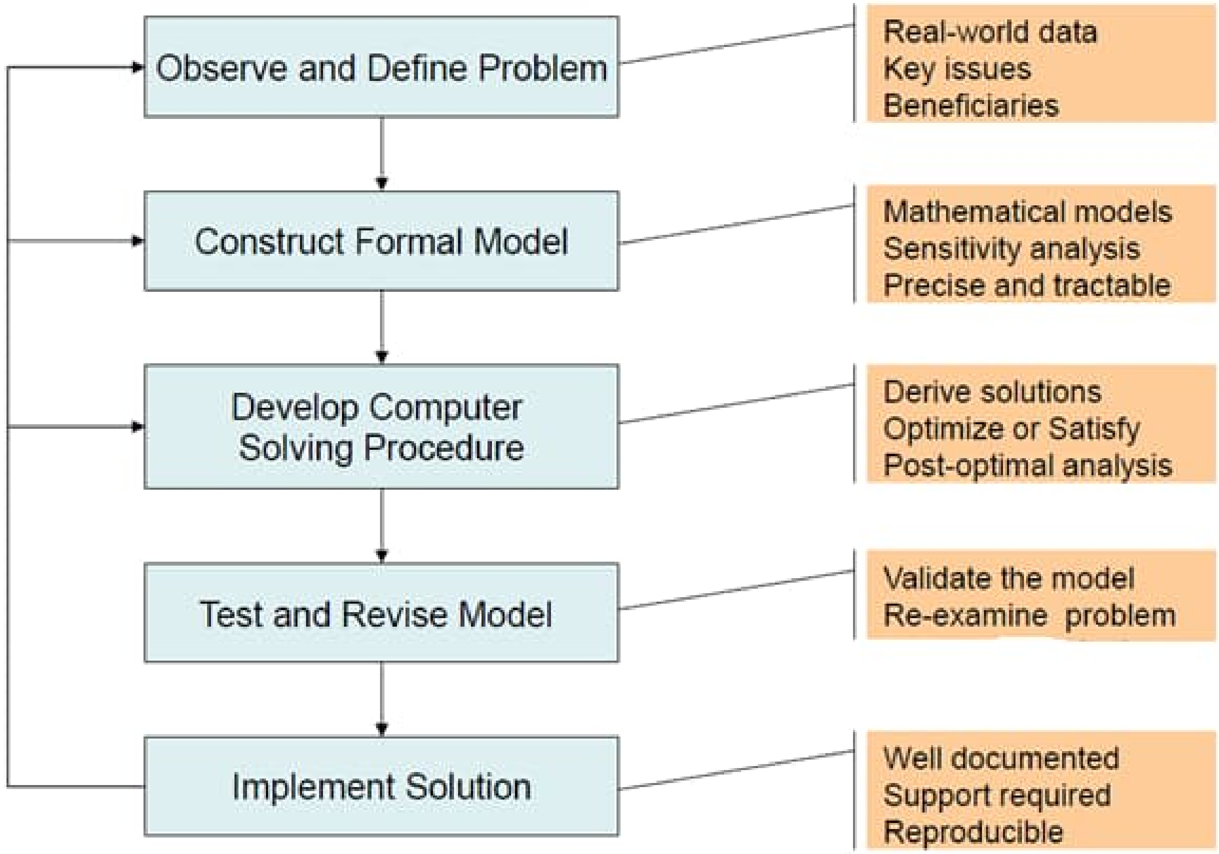
- Model validation;

5. Implement



- The success of the implementation phase depends a great deal upon the support of both top management and operating management.
- Kept management well informed good communications are key.

General framework for optimisation



Introduction to Linear Programming

- One specific type of optimisaiton, linear programming (LP), is used in all types of organizations to solve a wide variety of problems.
- LP is optimisation of a linear objective function, subject to linear equality and linear inequality constraints.
- Example: linear function, linear constraint.

Linear Programming: an overview

- Objectives of business decisions frequently involve **maximizing profit** or **minimizing costs.**
- Linear programming uses **linear algebraic relationships** to represent a firm's decisions, given a business **objective**, and resource **constraints**.
- Steps in application:
  1. Identify problem as solvable by linear programming.
  2. Formulate a mathematical model of the unstructured problem.
  3. Solve the model.

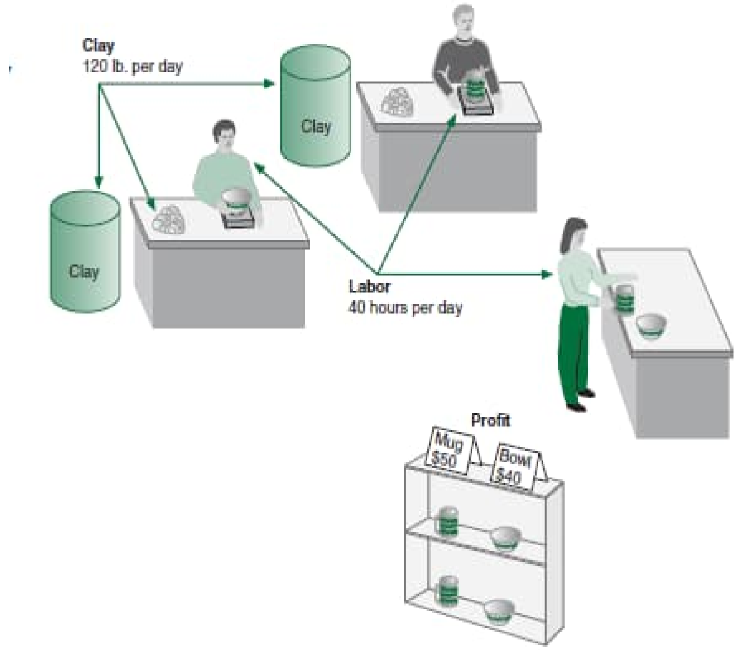
Model components

- **Decision variables** - mathematical symbols representing levels of activity by the organization.
- **Objective function** - a linear mathematical relationship describing an objective of the organization, in terms of decision variables - this function is to be maximized or minimized.
- **Constraints** - requirements or restrictions placed on the organization by the operating environment, stated in linear relationships of the decision variables.
- **Parameters** - numerical coefficients and constants used in the objective function and constraints.

LP modelling- an example

- **Resource Requirements**

product	Labor (Hr./Unit)	Clay (Lb./Unit)	Profit (£/Unit)
Bowl	1	4	40
Mug	2	3	50



- Product mix problem - Pottery Company
- How many bowls and mugs should be produced to maximize profits given labor and materials constraints?

Summary of model formulation steps

Step 1: Define the decision variables

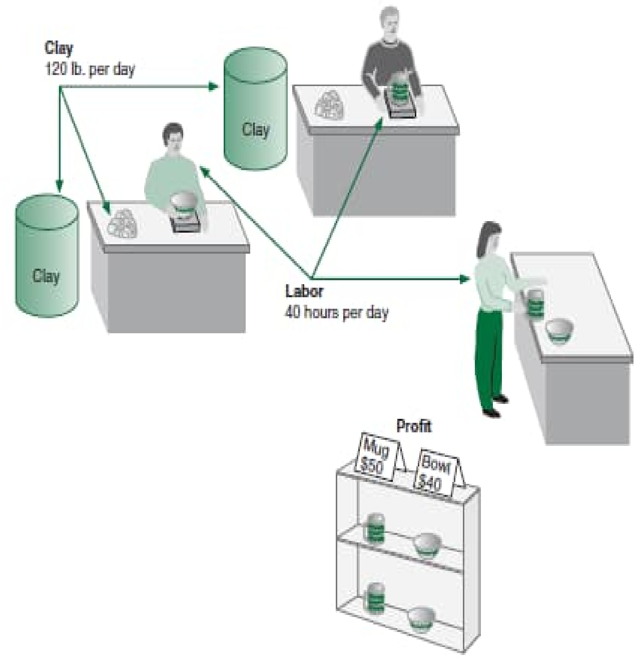
How many bowls and mugs to produce?

Step 2: Define the objective function

Maximize profit

Step 3: Define the constraints

The resources (clay and labor) available



LP model formulation

- *Define the decision variables*:

  x_1 = number of bowls to produce per day

  x_2 = number of mugs to produce per day
- *Describe the objective*: to maximise the total profit of all products:
- Maximize $Z = 40x_1 + 50x_2$
- *Describe the constraints*: $1x_1 + 2x_2 \leq 40$
- $4x_1 + 3x_2 \leq 120$

  $x_1 \geq 0; x_2 \geq 0$

LP model formulation

- Complete Linear Programming Model:

$$\max 40x_1 + 50x_2$$

$$s.t. 1x_1 + 2x_2 \leq 40$$

$$4x_2 + 3x_2 \leq 120$$

$$x_1, x_2 \geq 0$$

Feasible solution

- A **feasible solution** does not violate **any** of the constraints:
- Example:

$$x_1 = 5 \text{ bowls}$$

$$x_2 = 10 \text{ mugs}$$

$$Z = 40x_1 + 50x_2 = 700$$

- Labor constraint check: $1(5) + 2(10) = 25 \leq 40 \text{ hours}$
- Clay constraint check: $4(5) + 3(10) = 70 \leq 120 \text{ pounds}$

Infeasible solution

- An **infeasible solution** violates **at least one** of the constraints:
- Example:

$$x_1 = 10 \text{ bowls}$$

$$x_2 = 20 \text{ mugs}$$

$$Z = 40x_1 + 50x_2 = 1400$$

- Labor constraint check: $1(10) + 2(20) = 50 > 40$ hours

Graphical method of LP models

- Graphical method is limited to linear programming models containing **only two decision variables** (can be used with three variables but only with great difficulty).
- Graphical method provide **a picture of how** a solution for a linear programming problem is obtained.

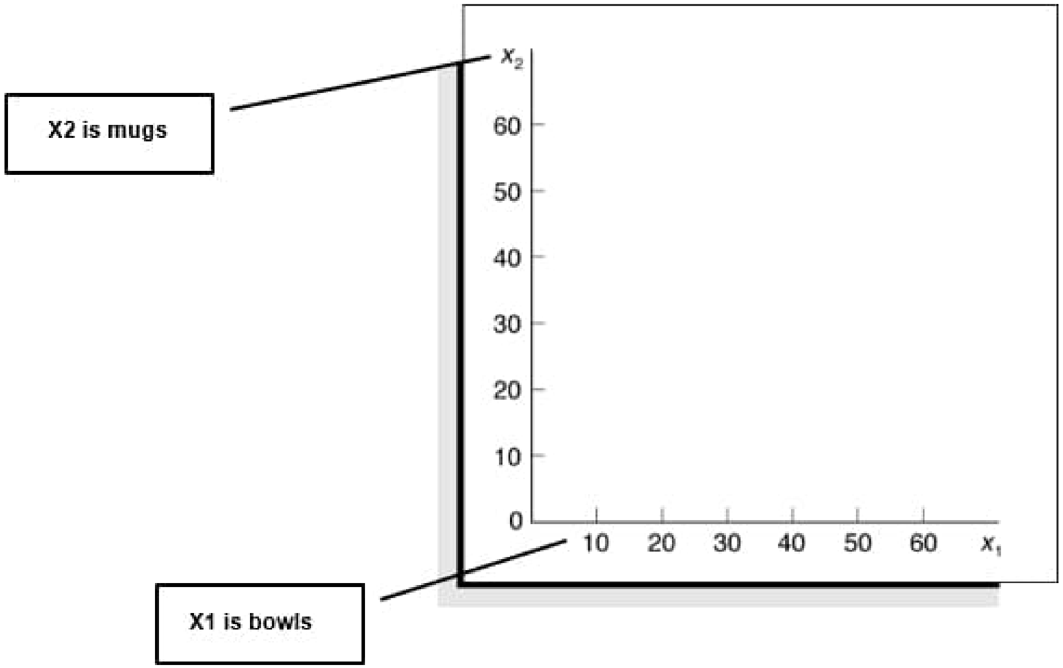
Coordinate axis

$$\max 40x_1 + 50x_2$$

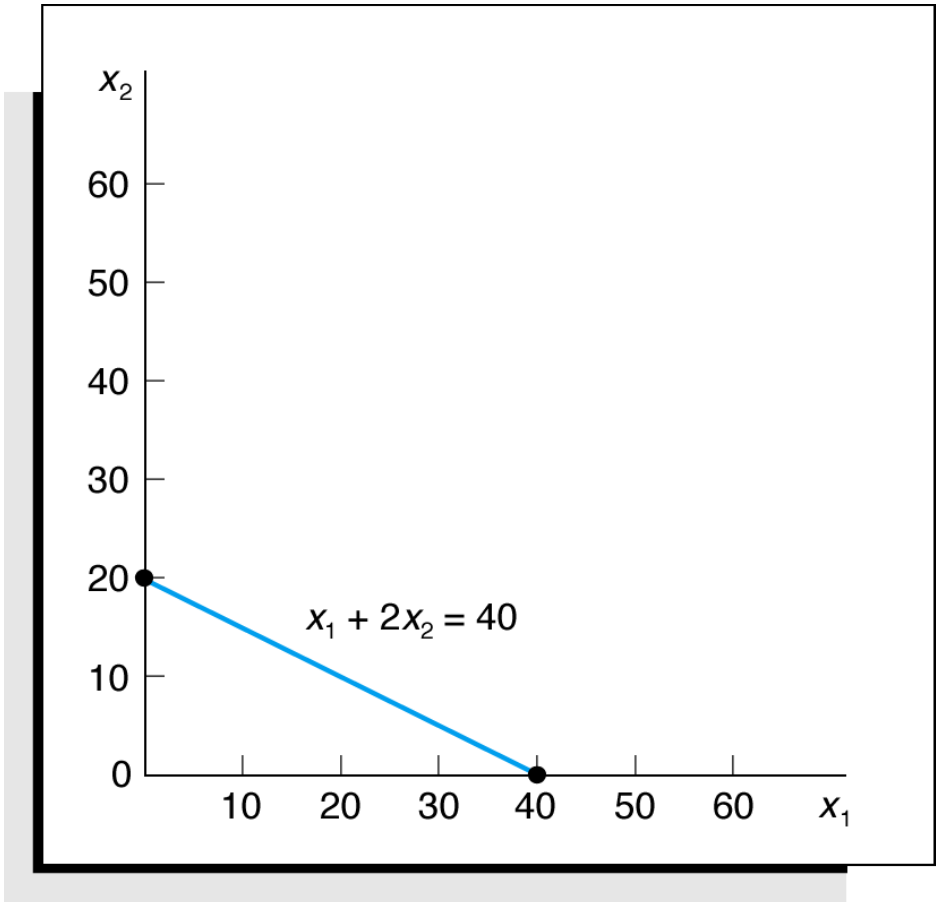
$$s.t.\ 1x_1 + 2x_2 \quad \leq 40$$

$$4x_2 + 3x_2 \leq 120$$

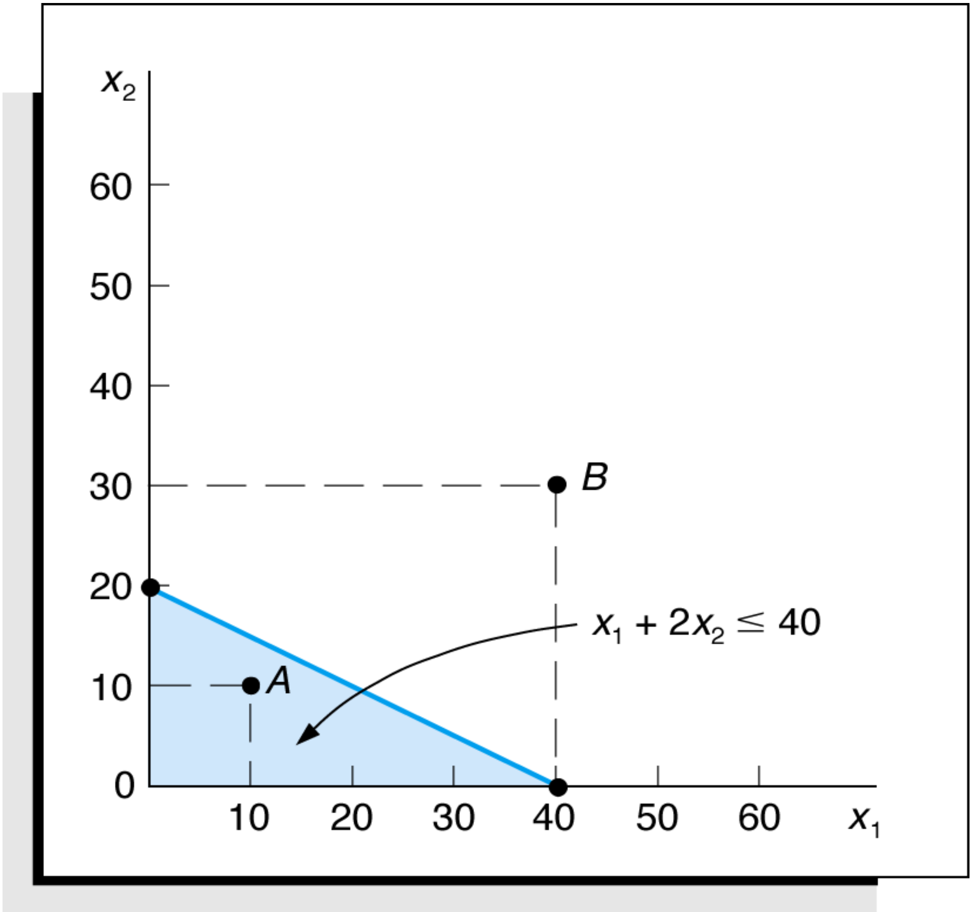
$$x_1, x_2 \geq 0$$



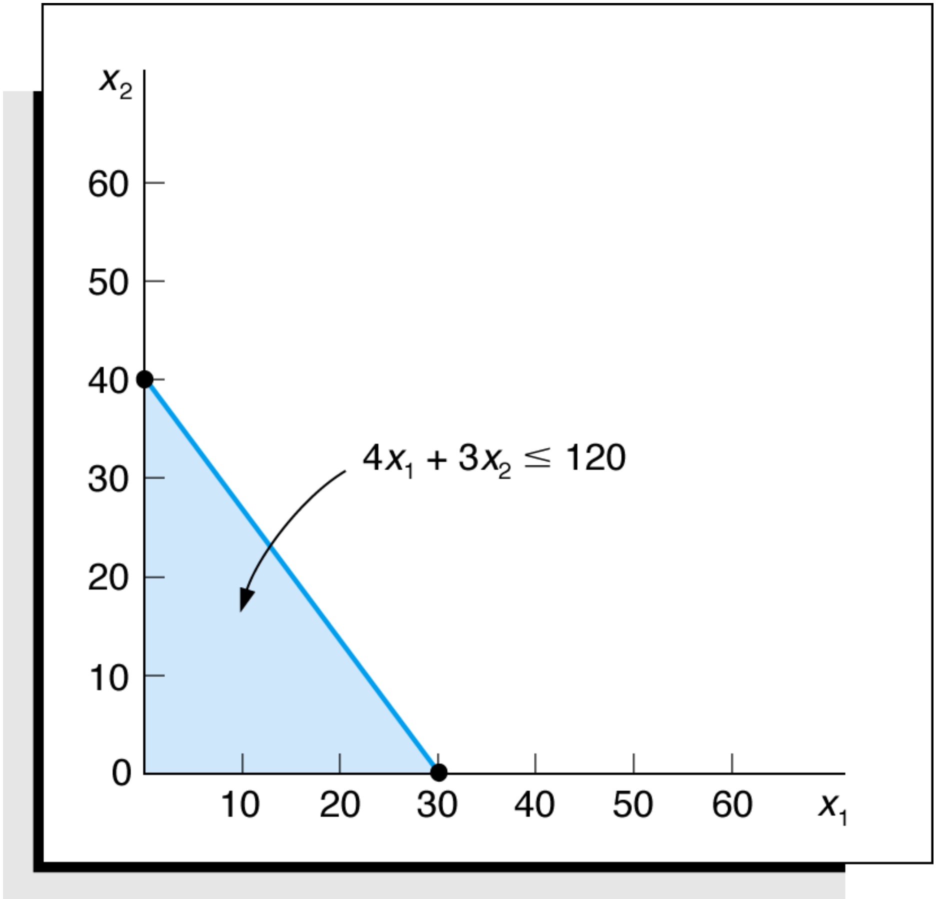
Labor constraint



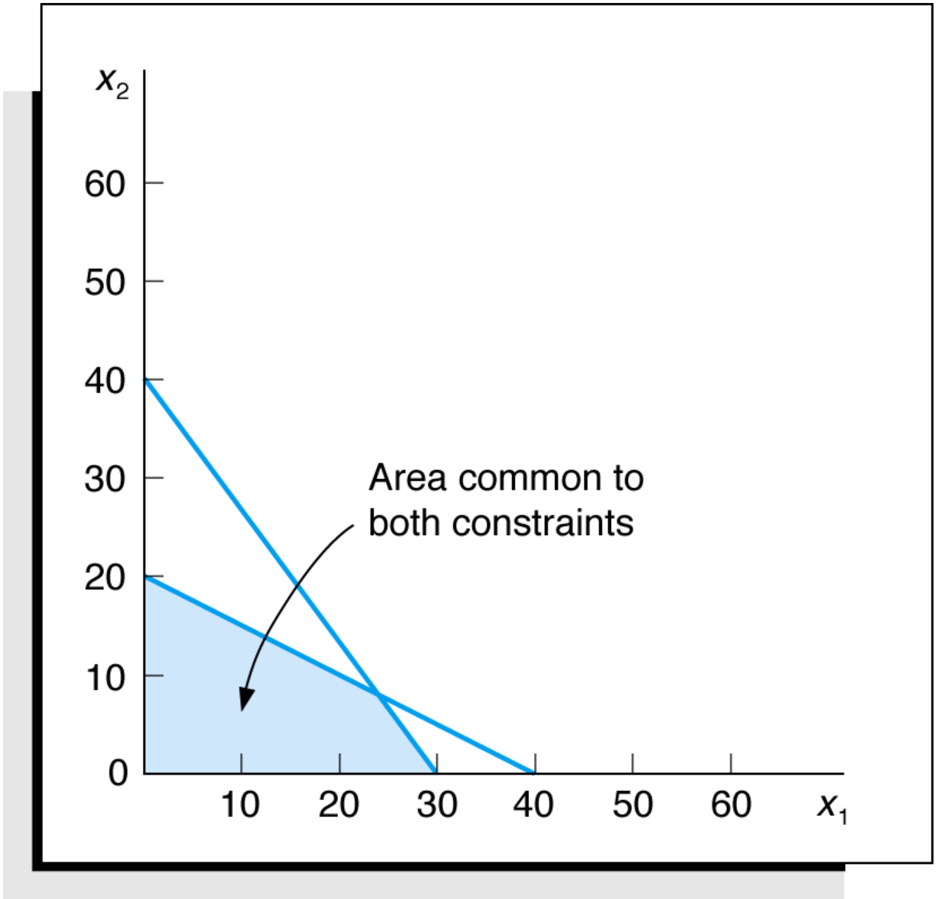
Labor constraint feasible area



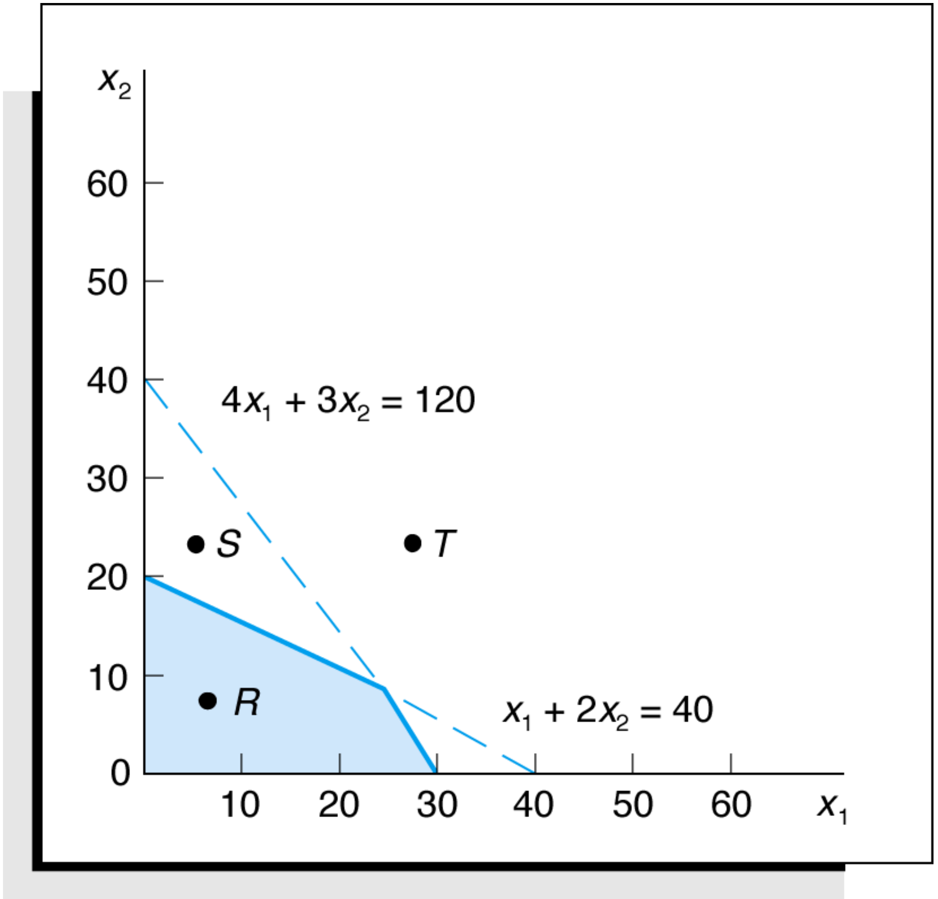
Clay constraint feasible area



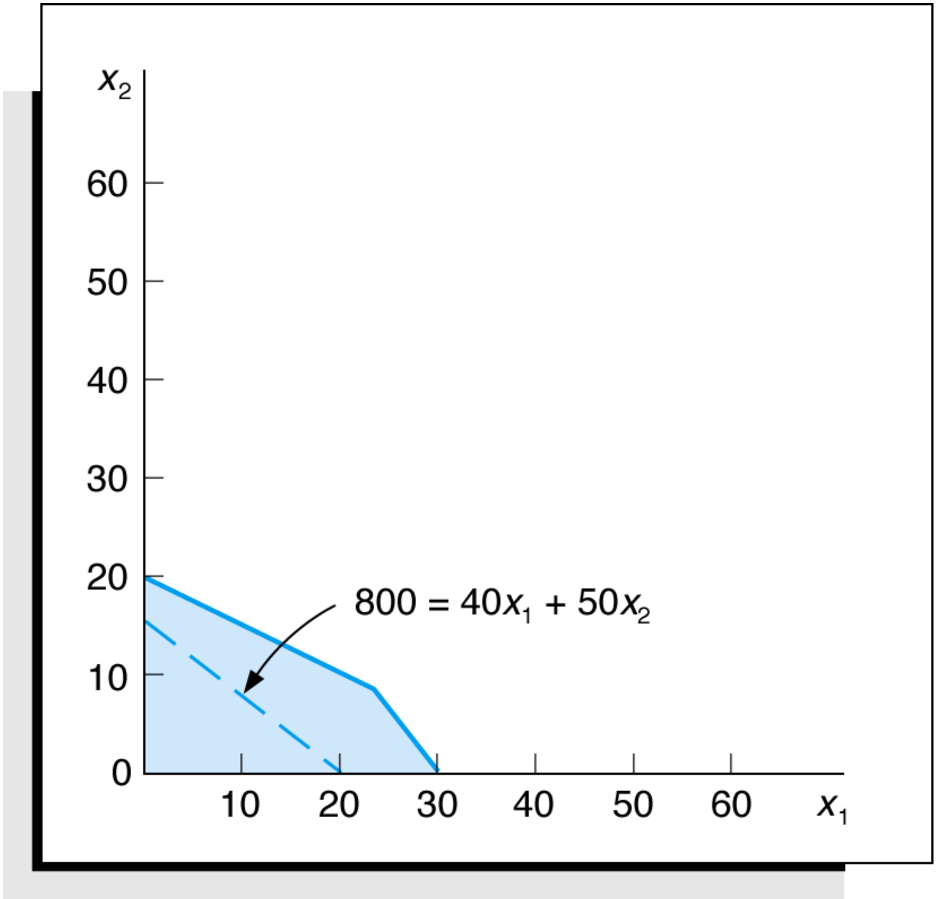
Feasible area for both constraints



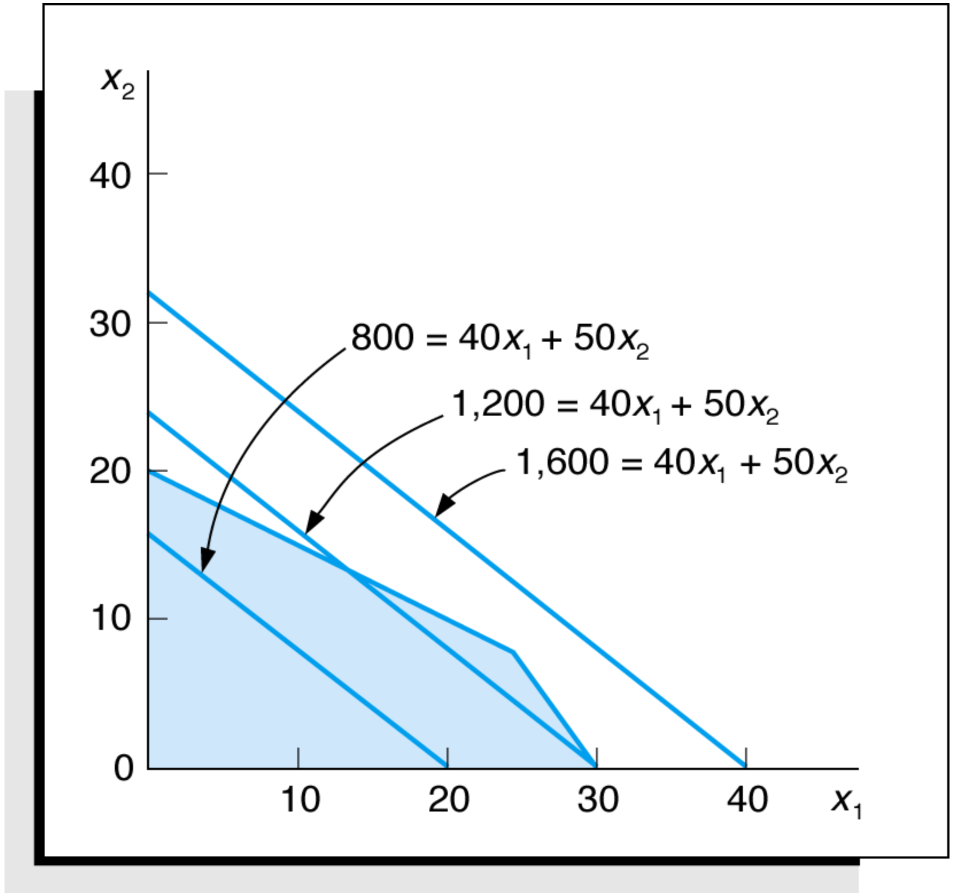
Feasible solution area



Objective function line with Z=800

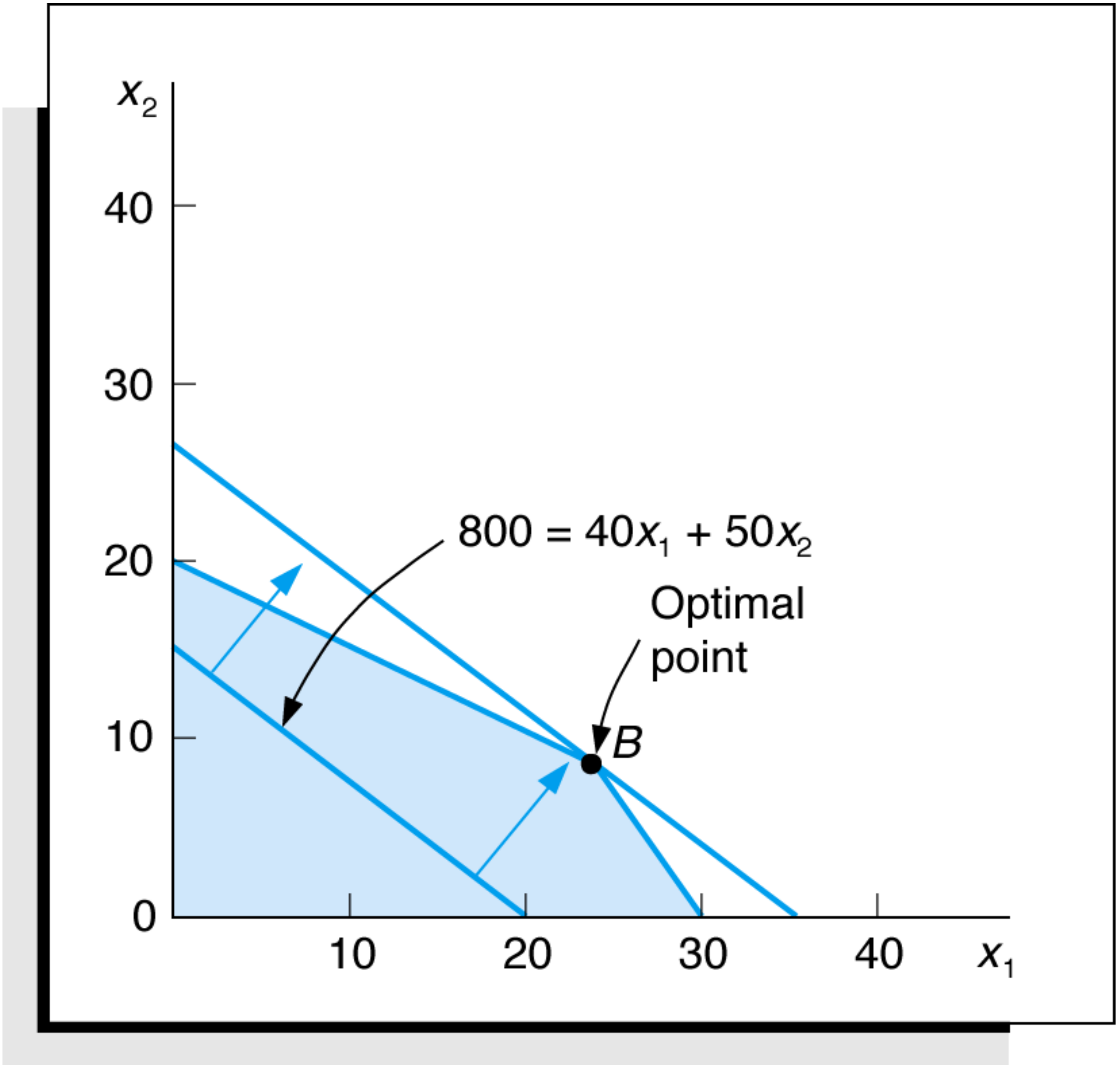


Alternative objective solution lines

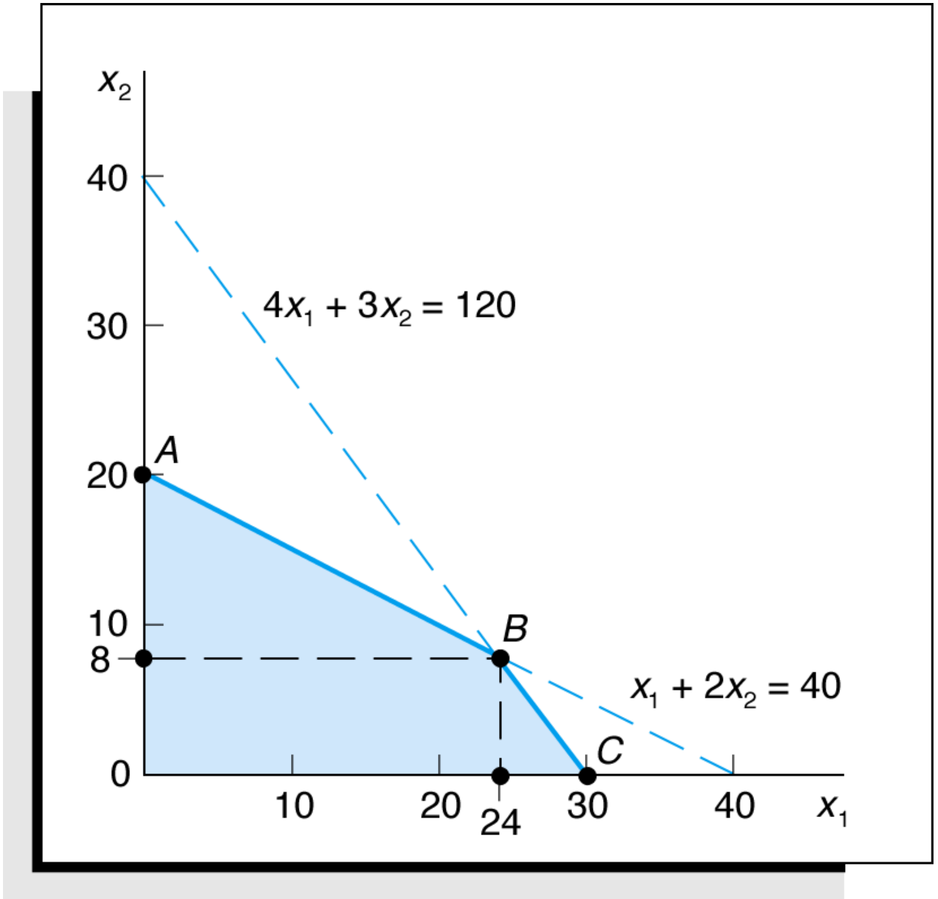


Optimal solution

- The optimal solution point is the last point the objective function touches as it leaves the feasible solution area.



Optimal solution coordinate

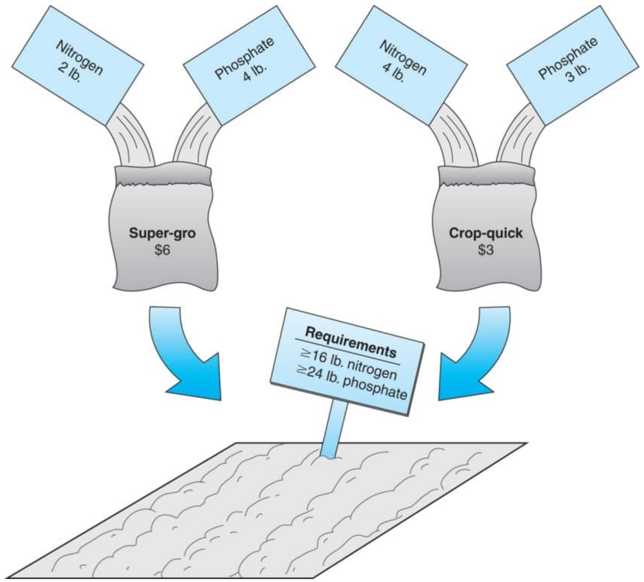


Graphical method for LP summarize

- 1. Draw a two-dimensional graph;
- 2. Draw lines representing the constraints;
- 3. Identify the feasible and infeasible regions;
- 4. Start with some solution in the feasible region and draw the line representing the objective function;
- 5. Find better values for the objective function within the feasible region;
- 6. Identify the direction that improves on the objective function;
- 7. Identify the line that contains the optimal solution.

A minimisation model example

- Two brands of fertilizer available – Super-gro, Crop-quick.
- Field requires at least 16 pounds of nitrogen and 24 pounds of phosphate.
- Super-gro costs £6 per bag, Crop-quick £3 per bag.
- Problem: How much of each brand to purchase to minimize total cost of fertilizer given following data ?



Chemical Contribution

Brand	Nitrogen (lb./bag)	Phosphate (lb./bag)
Super-gro	2	4
Crop-quick	4	3

LP model formulation

Decision Variables:

x_1 = bags of Super-gro

x_2 = bags of Crop-quick

The Objective Function:

Minimize $Z = 6x_1 + 3x_2$

Where: $6x_1$ = cost of bags of Super-Gro

$3x_2$ = cost of bags of Crop-Quick

Model Constraints:

$$2x_1 + 4x_2 \geq 16 \text{ lb} \ \ (\text{nitrogen constraint})$$

$$4x_1 + 3x_2 \geq 24 \text{ lb} \ \ (\text{phosphate constraint})$$

$$x_1,\ x_2 \geq 0 \ \ (\text{non-negativity constraint})$$

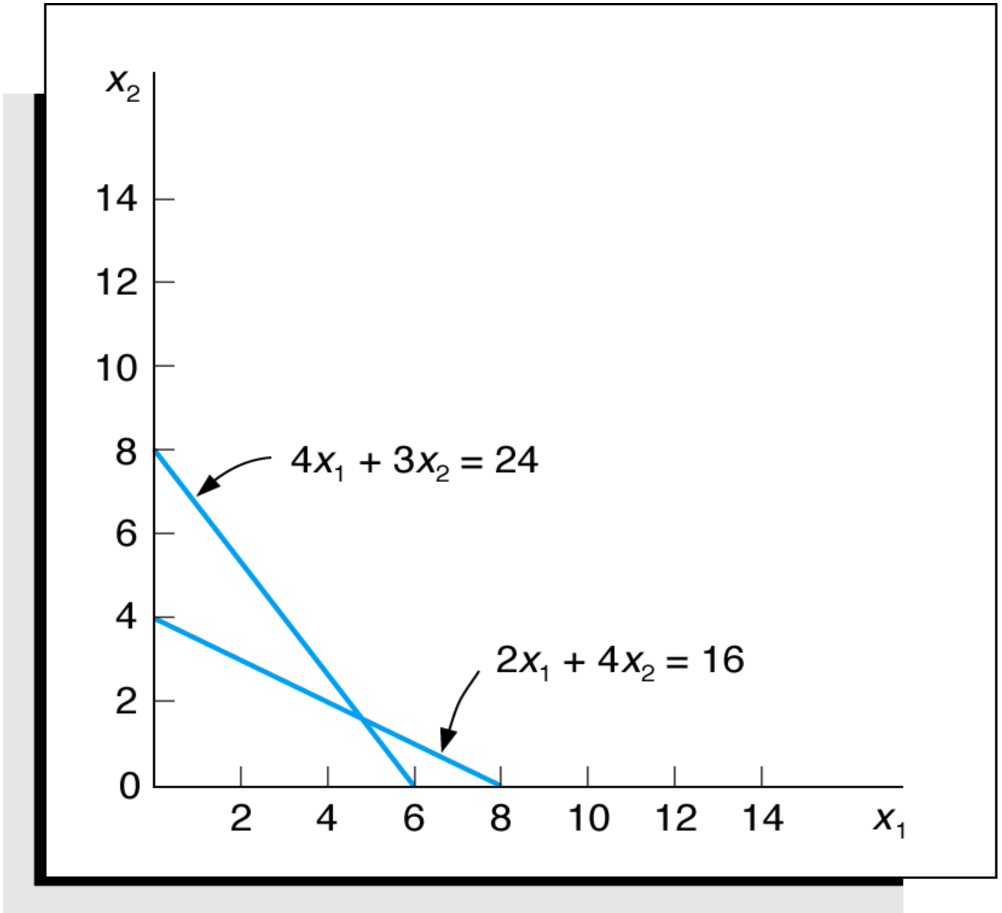
Constraint graph

$$\min 6x_1 + 3x_2$$

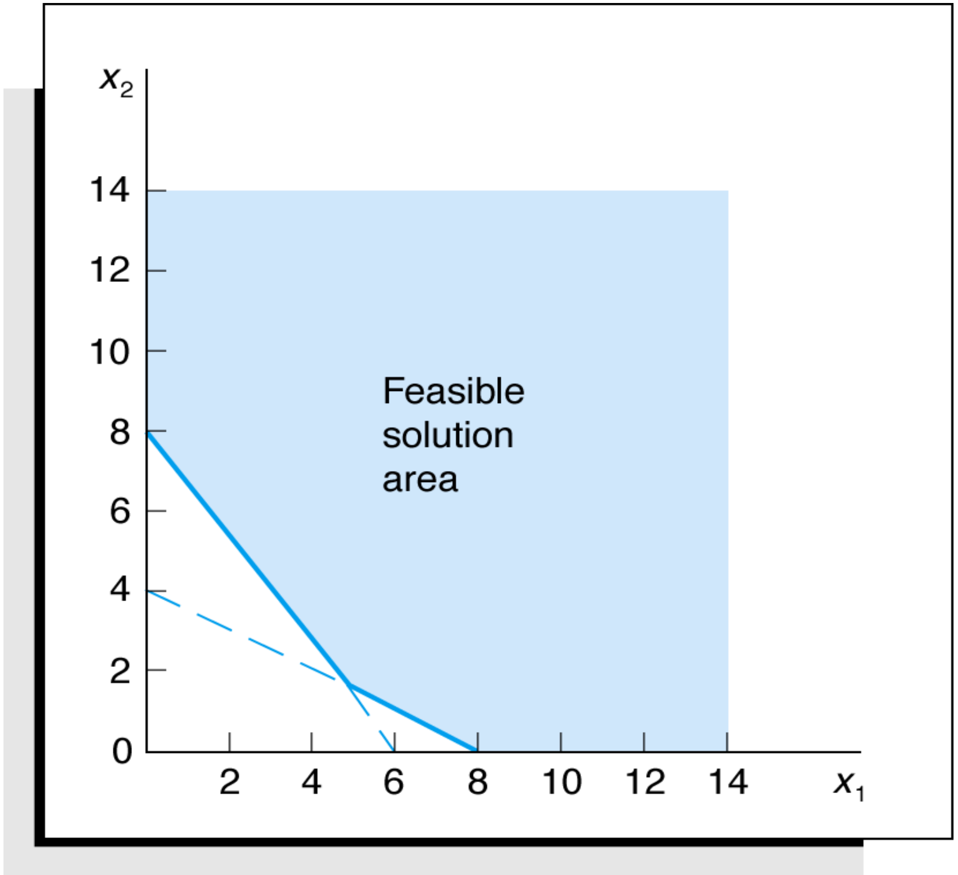
$$s.t. 2x_1 + 4x_2 \geq 16$$

$$4x_2 + 3x_2 \geq 24$$

$$x_1, x_2 \geq 0$$

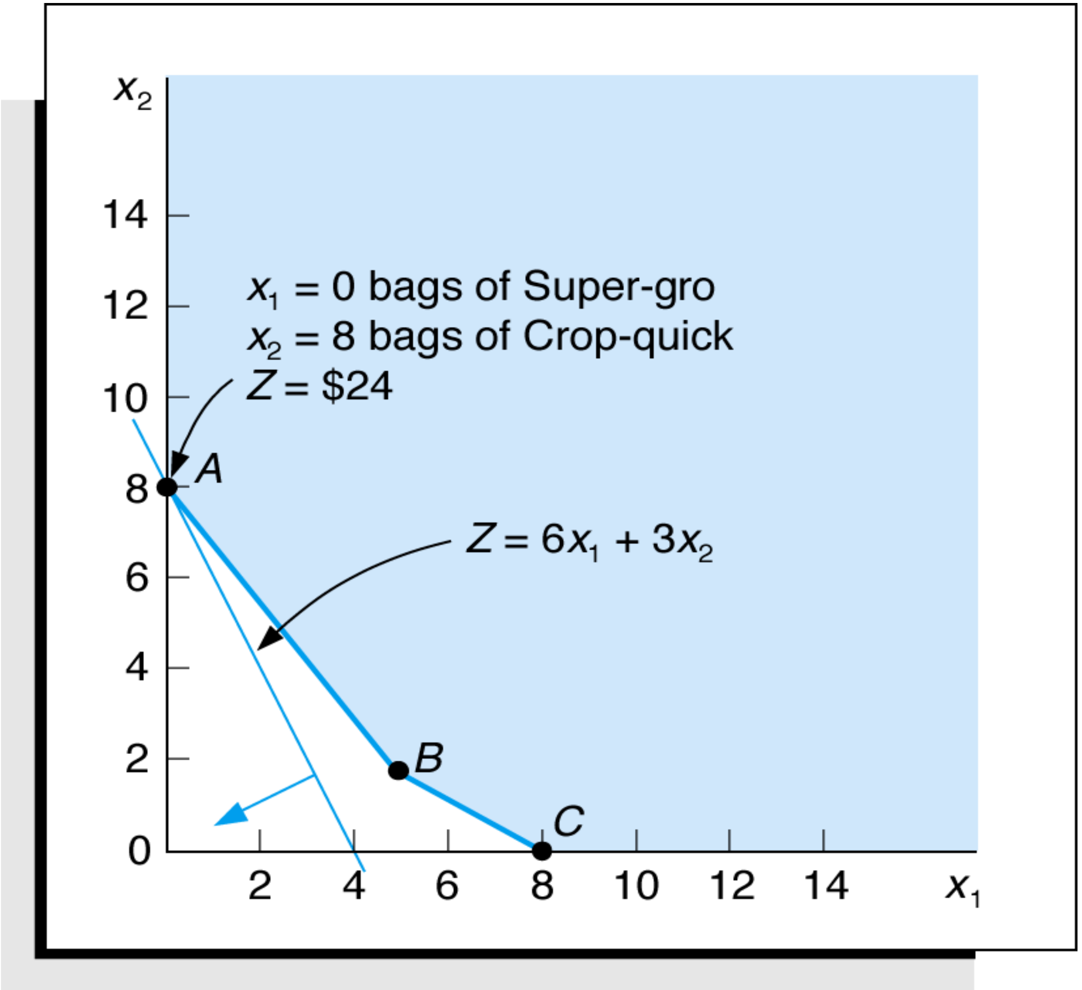


Feasible region



Optimal solution

- The optimal solution of a minimization problem is at the extreme point closest to the origin.



Special cases

Special types of problems include those with:

- Multiple optimal solutions
- Infeasible solutions
- Unbounded solutions

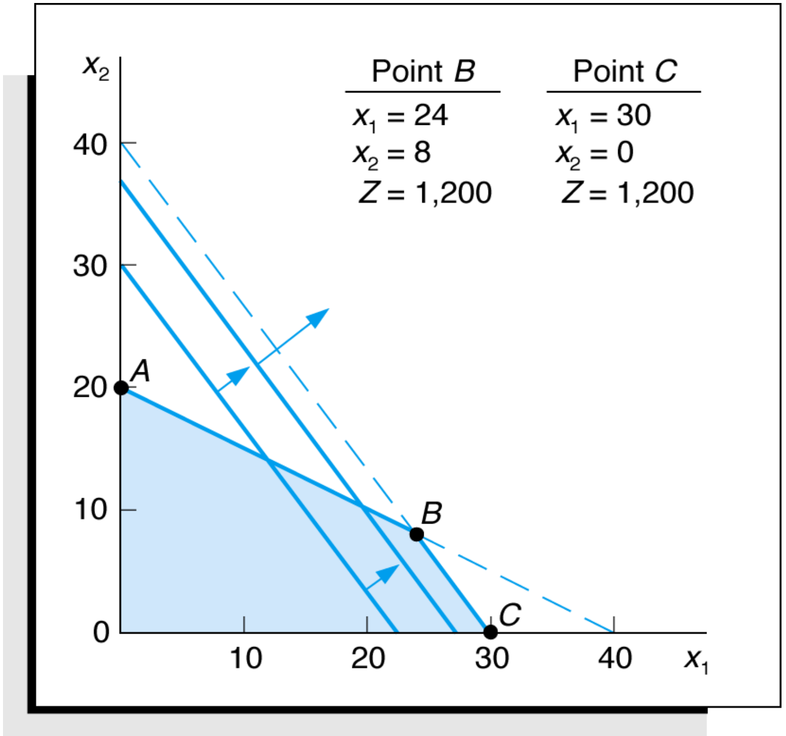
Multiple optimal solutions

$$\max 40x_1 + 30x_2$$

$$s.t.1x_1 + 2x_2 \leq 40$$

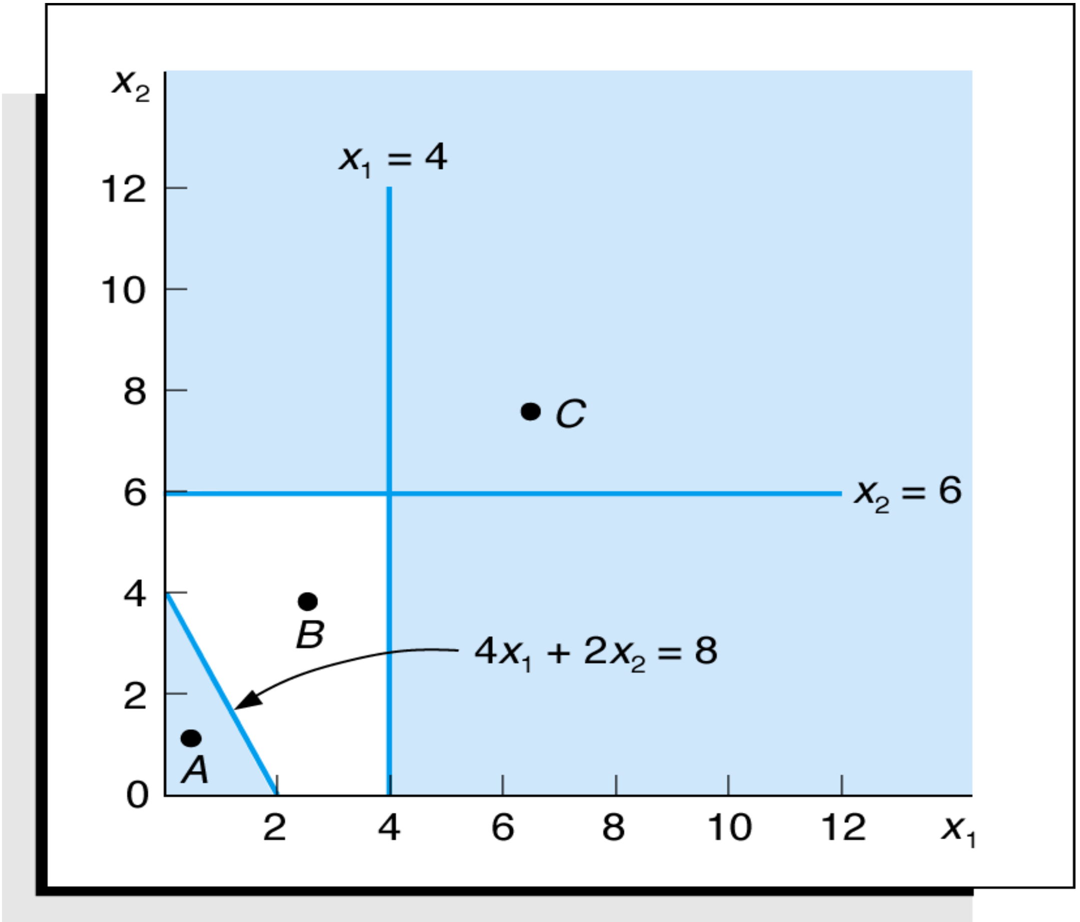
$$4x_2 + 3x_2 \leq 120$$

$$x_1,\ x_2 \geq 0$$



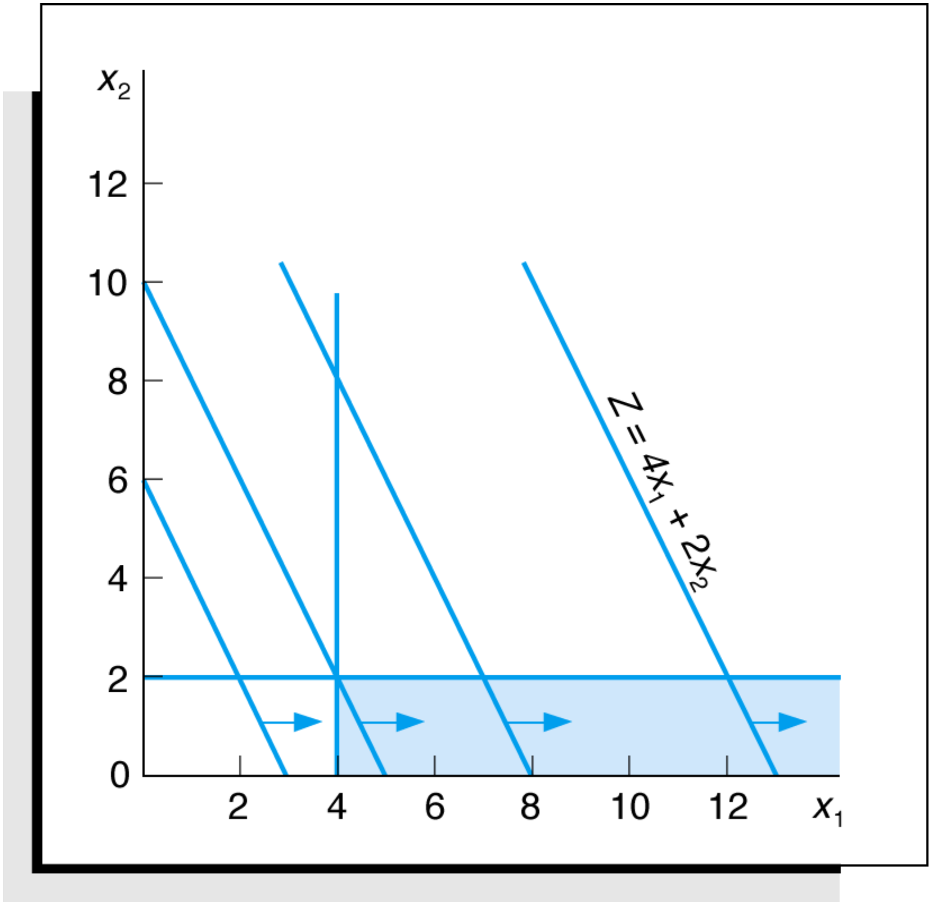
Infeasible problem

$$\max 5x_1 + 3x_2$$
$$s.t. 4x_1 + 2x_2 \leq 8$$
$$x_1 \geq 4$$
$$x_2 \geq 6$$
$$x_1,\ x_2 \geq 0$$



Unbounded problem

$$\max 4x_1 + x_2$$
$$s.t. x_1 \geq 4$$
$$x_2 \leq 2$$
$$x_1,\ x_2 \geq 0$$



Characteristics of LP problems

- A decision among alternative actions is required.
- The decision is represented in the model by **decision variables**.
- The problem encompasses a goal, expressed as an **objective function**, that the decision maker wants to achieve.
- Restrictions (represented by **constraints)** exist that limit the extent of achievement of the objective.
- The objective and constraints must be definable by **linear** mathematical functional relationships.

Put into practice

- Start with understanding and defining problem by reading the Case study “Assign regions to sales representatives”;

Questions?